STAFF REVIEW · H446 126 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. LDA (load).

2. The left subtree.

3. 24.

4. An input used by a subroutine.

I Do — source worked steps

1. #12 gives literal 12 — Predict the first step from the prompt.

2. 12 reads memory[12]=40 — Explain the intermediate result or decision.

3. @12 follows pointer to 7 — Using illustrative assembler conventions: immediate gives 12; direct reads 40; indirect follows memory[12]=40 then loads memory[40]=7.

We Do — guided CHECK

1. Base 80, index 3
Predict the first step from the prompt.
Effective address 83
Explain the intermediate result or decision.
Read memory[83]=19
Effective address = 80+3=83. It loads 19. Indexed addressing avoids writing a new absolute address for each element.

You Do — source mark schemes

1. Immediate uses supplied literal [2]; direct accesses memory at given address [2]; valid contrasting examples [2].

2. Direct =29 [3]. Indirect follows pointer 29, then loads 6 [3].

3. Effective address base + offset =54 [3], retrieves element stored there [2], supports array traversal [1], standard LMC has a restricted instruction set and does not natively specify general indexed addressing [2].

Exit Ticket — source answer

1. False. Direct uses the operand as a memory address; immediate uses a literal.